

Factor XIa Inhibitors for Secondary Prevention After Noncardioembolic Stroke

A Systematic Review and Meta-Analysis

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Abstract

Objectives

Patients with noncardioembolic ischemic stroke or transient ischemic attack (TIA) remain at substantial risk of recurrent ischemic events, despite guideline-recommended antithrombotic therapy. Factor XIa (FXIa) inhibitors may reduce thrombotic risk while preserving hemostasis.

Methods

We systematically searched MEDLINE and Scopus for randomized-controlled clinical trials (RCTs) comparing FXIa inhibitors with placebo in patients with acute noncardioembolic ischemic stroke or TIA. Primary efficacy outcome was first stroke (ischemic, hemorrhagic, or undefined), and primary safety outcome was first major bleeding event during follow-up. Risk ratios (RRs) with 95% CIs were pooled using random-effects models.

Results

Three RCTs recruiting 14,239 participants (6,927 assigned to FXIa inhibitors and 7,312 to placebo) were included. Effect estimates were largely driven by a single phase 3 trial, recruiting 12,327 participants. FXIa inhibitors reduced the risks for any stroke (RR = 0.75; 95% CI 0.66–0.84), ischemic stroke (RR = 0.74; 95% CI 0.66–0.84), and composite cardiovascular events (RR = 0.83; 95% CI 0.75–0.92), when compared with placebo. FXIa inhibitors did not increase the risks for major bleeding (RR = 1.12; 95% CI 0.87–1.44), hemorrhagic stroke, intracranial hemorrhage, any bleeding, or all-cause mortality.

Discussion

The addition of a FXIa inhibitor to standard antiplatelet therapy after an acute noncardioembolic ischemic stroke or TIA reduces the risk of stroke recurrence without evidence of an increased bleeding risk.

Introduction

Despite guideline-recommended antithrombotic strategies, patients after an ischemic stroke or TIA continue to face a substantial residual lifetime risk of stroke recurrence.¹ Randomized-controlled clinical trials (RCTs) not only established the benefit of short-term dual antiplatelet therapy after an ischemic stroke or TIA but also demonstrated an increased bleeding risk with prolonged dual antiplatelet therapy.² The combination of low-dose rivaroxaban with aspirin was found to significantly reduce the long-term cardiovascular risk of patients with atherosclerotic vascular disease and prior stroke but increase the risk for major bleeding.³

Factor XI emerged as a promising drug target for its key role in thrombosis and limited association with hemostasis.⁴ Based on converging clinical, epidemiologic, genetic, and experimental evidence suggesting a causal association between factor XI levels and ischemic stroke risk, trials have tested

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Supplementary Material

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the efficacy and safety of activated factor XI (FXIa) inhibitors for the prevention of recurrence after a noncardioembolic stroke or TIA.⁵

In the present systematic review and meta-analysis, we evaluate the efficacy and safety of FXIa inhibitors in secondary prevention after a noncardioembolic stroke or TIA.

Methods

The present systematic review and meta-analysis is reported according to the Preferred Reporting Items of Systematic Reviews and Meta-Analyses statement.⁶ The protocol is registered in PROSPERO.⁷

We searched MEDLINE and Scopus for published RCT testing FXIa inhibitors in patients with noncardioembolic stroke or TIA within 72 hours of onset. For phase 2 RCTs evaluating multiple doses, we analyzed only the dose that was tested forward in subsequent phase 3 RCTs. The search algorithm and study selection process are provided in the Supplementary Material (eFigure 1). Risk of bias for each eligible study was assessed with the Cochrane Collaboration risk of bias tool version 2.0, while the quality of summary evidence for every outcome of interest was evaluated using the Grading of Recommendations Assessment, Development, and Evaluation Working Group.⁸

The outcomes of interest and the statistical analysis are outlined in the eMethods.

Results

We included 3 RCTs including a total of 14,239 patients (6,609 randomized to asundexian 50 mg daily, 318 to milvexian 25 mg twice daily, and 7,312 to placebo).^{9,10,11} Inclusion criteria were similar between RCTs, recruiting patients within 48 or 72 hours from the onset of a noncardioembolic stroke or TIA (Table 1), while baseline characteristics in the FXIa groups were comparable across RCTs (eTable 1). All participants in AXIOMATIC Secondary Stroke Prevention (SSP) were on dual antiplatelet therapy at the time of randomization, while the respective percentages for PACIFIC-Stroke and OCEANIC-Stroke were 43% and 63%. Risk of bias was low in all RCTs (eFigure 2).

Participants randomized to a FXIa inhibitor had lower risk for any stroke (risk ratio [RR] = 0.75, 95% CI 0.66–0.84; number needed to treat = 48; Figure, A), ischemic stroke (RR = 0.74, 95% CI 0.66–0.84; eFigure 3), and the composite cardiovascular outcome (RR = 0.83, 95% CI 0.75–0.92; eFigure 4) compared with those randomized to placebo. The risk for major bleeding was comparable between participants receiving FXIa inhibitor or placebo (RR = 1.12, 95% CI 0.87–1.44; Figure, B). There was no significant difference in the risks for cardiovascular death (eFigure 5), hemorrhagic

stroke (eFigure 6), intracranial hemorrhage (eFigure 7), any bleeding (eFigure 8), all-cause mortality (eFigure 9), or TIA (eFigure 10) between the FXIa and control arms (Table 2).

Discussion

Our aggregate data meta-analysis showed that adding a FXIa inhibitor to standard antiplatelet therapy after an acute noncardioembolic ischemic stroke or TIA significantly reduces the risks for incident stroke, ischemic stroke, and composite cardiovascular outcome, when compared with placebo. These benefits were achieved without increase in major bleeding, hemorrhagic stroke, intracranial hemorrhage, or all-cause mortality.

Our findings on the safety of FXIa inhibitors in this meta-analysis are in line with a pooled analysis of phase 2 RCTs testing different doses of asundexian in 4,164 patients with atrial fibrillation, recent acute myocardial infarction, or stroke suggesting less bleeding risk with asundexian when compared with apixaban, and similar risk for major or any bleeding when asundexian was compared with placebo.¹² These results are also in line with a previous aggregate data meta-analysis of 4 phase II dose-finding RCTs that provided reassuring evidence regarding the safety of FXIa inhibitors.¹³ Moreover, in a secondary analysis of 1,507 participants from PACIFIC-Stroke with baseline and follow-up MRIs, there were numerically less incident cerebral microbleeds in patients with microbleeds assigned to asundexian 50 mg daily compared with placebo.¹⁴

Certain limitations need to be taken into account for the interpretation of the findings of the meta-analysis. First, it needs to be highlighted that effect estimates were largely driven by a single phase 3 trial,¹⁰ which has disproportionately influenced pooled results. In this context, it is important to distinguish that phase II studies are primarily exploratory and designed to generate preliminary efficacy signals, whereas phase III trials provide confirmatory evidence of clinical benefit and safety. We detected no heterogeneity across the 3 RCTs for both the primary efficacy and safety end points. In addition, there was no or limited heterogeneity ($I^2 < 40\%$) across the different trials for all the secondary end points with the exception of incident TIA ($I^2 = 51\%$). However, owing to the relatively small number of included studies, the robustness of between-study heterogeneity assessments is limited and should be interpreted cautiously. Second, it should be noted that most included participants were treated with asundexian, limiting the generalizability of reported findings to other FXIa inhibitors. Although both asundexian and milvexian target the same pathway, they are distinct compounds with potentially relevant pharmacodynamic and pharmacokinetic differences. It also should be acknowledged that the follow-up duration of included trials varied substantially, as presented in Table 1. AXIOMATIC-SSP followed patients for 90 days, providing limited data on long-term efficacy and

Table 1 Overview of Included Trials

	AXIOMATIC-SSP ⁹	OCEANIC-Stroke ¹⁰	PACIFIC-Stroke ¹¹
Population			
Age	≥40 y	≥18 y	≥45 y
Type of event	Noncardioembolic, nonlacunar IS or high-risk TIA	Noncardioembolic or high-risk TIA	Noncardioembolic IS
Event severity	NIHSS ≤ 7 ^a points for IS ABCD2 ≥ 6 for TIA	NIHSS ≤ 15 for IS ABCD2 ≥ 6 for TIA	NIHSS ≤ 15 ^b
Time from onset	≤48 h	≤72 h	≤48 h
Other criteria	Visible intracranial or extracranial atherosclerosis in an artery considered to be anatomically related to the index event	History of atherosclerosis (CAD, PVD, carotid stenosis ≥50%) Nonlacunar IS Imaging evidence of cerebrovascular atherosclerosis of any degree involving extracranial or intracranial vessels or the aortic arch	None
Intervention	Milvexian 25 mg OD Milvexian 25 mg BID ^c Milvexian 50 mg BID Milvexian 100 mg BID Milvexian 200 mg BID	Asundexian 50 mg OD	Asundexian 10 mg OD Asundexian 20 mg OD Asundexian 50 mg OD ^c
Comparator	Matching placebo	Matching placebo	Matching placebo
Primary outcome	IS or CBI (on MRI) within 90 d	Incident ischemic stroke over a median follow-up of 567 d	IS or CBI (on MRI) within 26 wk

Abbreviations: ABCD2 = age, blood pressure, clinical features, duration of TIA and presence of diabetes; BID = twice daily; CAD = coronary artery disease; CBI = covert brain infarct; IS = ischemic stroke; NIHSS = NIH Stroke Scale; OD = once daily; PVD = peripheral vascular disease.

^aInitial phase NIHSS ≤5.

^bInitial phase NIHSS ≤7.

^cDose evaluated in the meta-analysis.

safety of milvexian,⁹ when compared with PACIFIC-Stroke and OCEANIC-Stroke evaluating asundexian for a median duration of 11 months and 19 months, respectively.^{10,11} In

a post hoc sensitivity analysis restricted to PACIFIC-Stroke and OCEANIC-Stroke, we found similar reduction in the risk of stroke recurrence with asundexian 50 mg once daily

Figure Risks for (A) Any Stroke or (B) Major Bleeding During Follow-Up Between Factor Xla Inhibitors and Placebo Arms

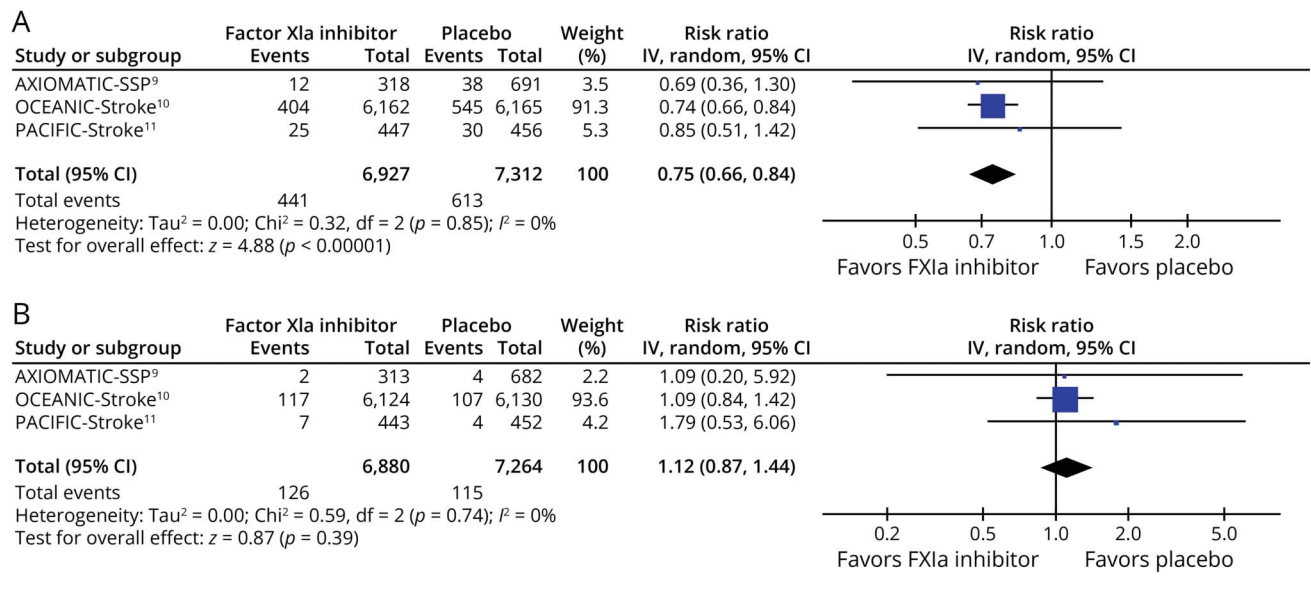


Table 2 Overview of the Efficacy and Safety End Points

Outcomes	No. of pts (studies)	Certainty of the evidence	RR (95% CI)	Anticipated absolute effects	
				Risk with placebo	Risk difference with factor XIa inhibitor
Efficacy end points					
Any stroke	14,239 (3 RCTs)	⊕⊕⊕⊕⊕ High	RR 0.75 (0.66–0.84)	84 per 1,000	21 fewer per 1,000 (from 29 fewer to 13 fewer)
Ischemic stroke	14,239 (3 RCTs)	⊕⊕⊕⊕ High	RR 0.74 (0.66–0.84)	80 per 1,000	21 fewer per 1,000 (from 27 fewer to 13 fewer)
Composite CV outcome	14,239 (3 RCTs)	⊕⊕⊕⊕ Moderate ^a	RR 0.83 (0.75–0.92)	104 per 1,000	18 fewer per 1,000 (from 26 fewer to 8 fewer)
CV death	13,322 (2 RCTs)	⊕⊕⊕⊕ Low ^b	RR 1.24 (0.30–5.11)	24 per 1,000	6 more per 1,000 (from 17 fewer to 98 more)
Safety end points					
Major bleeding	14,144 (3 RCTs)	⊕⊕⊕⊕ Moderate ^c	RR 1.12 (0.87–1.44)	16 per 1,000	2 more per 1,000 (from 2 fewer to 7 more)
Hemorrhagic stroke	13,263 (2 RCTs)	⊕⊕⊕⊕ Low ^b	RR 0.65 (0.32–1.31)	3 per 1,000	1 fewer per 1,000 (from 2 fewer to 1 more)
Intracranial hemorrhage	14,144 (3 RCTs)	⊕⊕⊕⊕ Low ^b	RR 1.16 (0.75–1.79)	5 per 1,000	1 more per 1,000 (from 1 fewer to 4 more)
Any bleeding	14,144 (3 RCTs)	⊕⊕⊕⊕ Moderate ^c	RR 0.98 (0.88–1.08)	99 per 1,000	2 fewer per 1,000 (from 12 fewer to 8 more)
All-cause mortality	14,239 (3 RCTs)	⊕⊕⊕⊕ Moderate ^c	RR 1.04 (0.82–1.31)	37 per 1,000	1 more per 1,000 (from 7 fewer to 11 more)
Exploratory end points					
Transient ischemic attack	14,225 (3 RCTs)	⊕⊕⊕⊕ Very low ^{b,d}	RR 0.68 (0.31–1.46)	22 per 1,000	7 fewer per 1,000 (from 15 fewer to 10 more)

Abbreviations: CV = cardiovascular; RCT = randomized clinical trial; RR = risk ratio.
^aHeterogeneity in the outcome definition across trials.
^bCI is very wide, including both important benefit and harm.
^cCI is relatively wide, including both benefit and harm.
^dPresence of moderate heterogeneity across effect estimates.

when compared with placebo (RR = 0.75, 95% CI 0.66–0.84; eFigure 11). LIBREXIA-STROKE (NCT05702034) is an ongoing phase 3 RCT assessing the safety and efficacy of milvexian 25 mg twice daily for secondary prevention after a non-cardioembolic stroke. Third, the results of this meta-analysis are only applicable to patient populations recruited in trials, and therefore, the utility of FXIa inhibition in patients with non-cardioembolic stroke or TIA presenting beyond 72 hours from symptoms onset, as well as in patients with cardioembolic stroke, remains uncertain. Pooled analyses of subgroups stratified by index stroke subtype, age, sex, and severity were not feasible because these data were reported in only one of the included studies.¹⁰ However, we found similar estimates in the risk of stroke recurrence in post hoc exploratory analyses including only patients from PACIFIC-Stroke and OCEANIC-Stroke with the index event attributed to large-artery atherosclerotic disease (RR = 0.80, 95% CI 0.67–0.95; eFigure 12) or participants from all studies with imaging evidence of any extracranial or intracranial atherosclerosis, whether causally linked with the index event (RR = 0.74, 95% CI 0.66–0.85; eFigure 13). Finally, it should be noted that the composite cardiovascular outcome

definition varied among trials because vascular death was used in OCEANIC-Stroke and PACIFIC-Stroke trials, but all-cause death was included in AXIOMATIC-SSP.

Current evidence suggests a net benefit for FXIa inhibitors when added to standard antiplatelet therapy after a non-cardioembolic ischemic stroke or TIA, representing an important advancement in secondary stroke prevention.

Author Contributions

A.H. Katsanos: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A. Shoamanesh: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. L. Palaodimou: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. M. Sharma: drafting/revision of the manuscript for content, including medical writing for content; major role in the

acquisition of data; analysis or interpretation of data. G. Tsivgoulis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

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